Mathematics Exercises

2026

ESQC mathematics exercises

2026 edition

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Preface

This is a collection of mathematics exercises for the European Summerschool in Quantum Chemistry (ESQC).

Topics needed (List from Lucas)

If an exercise is provided, please cross it out below.

- Second quantization
  - ☐ Binomial expansion
  - ☒ binomial coefficients
  - ☒ Pascal's triangle
  - ☐ Cauchy product of series
  - ☐ Proof by induction (or direct)
  - ☒ Handling summation signs (also in combination with Kronecker deltas)
  - ☐ Definition of exp as a power series
  - ☐ Partial derivatives of vectors (CI eigenvalue problem)
  - ☐ Finding eigenvalues and eigenvectors of a 2x2 matrix.
- SCF
  - ☒ Matrix algebra: matrix multiplication
  - ☐ unitary transformation
- DFT
  - ☐ functional derivatives
  - ☐ Transformation formula for multidimensional integrals (calculating Jacobi determinant for transformation from one set of coordinates to another).
- Response theory
  - ☐ BCH expansion, manipulation of exponentials of operators.

- ☐ Exponential of antihermitian operator is unitary (and maybe also the opposite: Every unitary can be written as an exponential of an antihermitian operator).
- ☐ Poles and residues of complex functions
- ☐ Fourier transform
- ☐ exponential form of the sin and cos functions
- ☐ Euler's formula for complex exponentials
- ☐ Partial derivatives with respect to dependent variables (In this case: the parameters Pn and their complex conjugates Pn?)

- Relativity
  - ☒ Levi-Civita symbol
  - ☐ Roots of a 3rd order polynomial
- Coupled cluster
  - ☐ Taylor expansion
- Molecular properties
  - ☐ Writing cross product in terms of Levi-Civita symbols
  - ☐ Evaluating commutator of momentum operator with a local operator (like vector potential)
  - ☐ Lagrange multiplier method
  - ☐ Total and partial derivatives
- Multiconfigurational methods
  - ☐ resolution of the identity, projection operators

Note

Current levels and recommendation values coded in automatic generation of list script:

```
vels = ["beginner", "intermediate", "advanced"]

recommendations = ["math", "topic"]

# Topics and their descriptions, the latter used for writing the recomm
# exercises per topic.
topics = {
```

```
    "coupled-cluster": "Coupled-cluster theory",
    "second-quantization": "Second quantization",
    "scf": "Self-consistent field theory",
    "dft": "Density functional theory",
    "response-theory": "Response theory",
    "relativity": "Relativistic quantum chemistry",
    "molecular-properties": "Molecular properties",
    "multiconfig-methods": "Multiconfigurational methods",
}
```

Part I

Exercise recommendations

Recommended exercises for the first day

This chapter contains recommended exercises for the mathematics tutorial on the first day of the school. These recommended exercises are considered especially useful.

We provide also a list of recommended exercises for the 'beginners'. If you feel that you need training in the *absolute basics* of the mathematics used in quantum chemistry, we recommend that you start here.

In the next chapter, we provide lists of recommended exercises for each quantum chemistry topic taught at the school.

Beginner

- Exercise 1.1 — Days of the week
- Exercise 1.10 — Classically allowed region for Morse potential
- Exercise 3.1 — Very basic sums
- Exercise 3.2 — Simplifying as a sum
- Exercise 3.4 — Reconstruct a vector
- Exercise 3.5 — Summing in a matrix–vector product
- Exercise 3.6 — The summation index is a dummy index
- Exercise 3.7 — Linearity of sums
- Exercise 3.8 — When something does not depend on the summation index
- Exercise 3.9 — Free and summed indices
- Exercise 3.10 — Shifting the summation index
- Exercise 3.11 — The Kronecker delta: basic exercise
- Exercise 4.1 — Basic factorials
- Exercise 5.1 — Basic binomial coefficients
- Exercise 5.2 — Pascal's triangle
- Exercise 6.1 — Basic vector operations
- Exercise 6.2 — Matrix-vector products
- Exercise 6.3 — Matrix-matrix products
- Exercise 6.4 — Linear independence and basis

- Exercise 6.5 — Gaussian elimination and vector decomposition
- **?@exr-general-vector-spaces-recipes** — Cooking with vectors 1
- **?@exr-bra-ket-column-vectors** — Bra-ket notation and column vectors

Intermediate

- Exercise 1.2 — Five first natural numbers
- Exercise 1.8 — De Morgan's laws
- Exercise 1.11 — Molecules as sets
- Exercise 2.1 — Countability of the integers
- Exercise 3.3 — Estimating $\sqrt{e}$ via a Taylor series
- Exercise 4.2 — Lunch line
- Exercise 6.6 — Office coordinates
- Exercise 6.7 — Boat sailing on the ocean
- Exercise 6.8 — An exercise by Robert Beezer
- Exercise 6.9 — Compute the action of a matrix on vectors
- **?@exr-general-vector-spaces-meaningful-recipes** — Cooking with vectors 2
- **?@exr-gaussian-elimination-compute-overlap-matrix-basis-basis-defined** — Gaussian elimination and overlap matrix
- **?@exr-polynomial-vector-spaces-set-all-polynomials-degree-less-than**
- **?@exr-polynomial-vector-spaces-differentiation-operator-i-e-indeed-linear**
- **?@exr-metric-spaces-metric-spaces**
- **?@exr-visualizing-functions-path-defined-sketch-curve-traced-path**
- **?@exr-visualizing-functions-sketch-level-curves**
- **?@exr-continuity-and-differentiability-given-does-limit-not-exist-why**
- **?@exr-continuity-and-differentiability-marsden-tromba-does-exist-if-you**
- **?@exr-complex-arithmetic-compute-real-imaginary-parts-functions**
- **?@exr-complex-arithmetic-complex-numbers-can-written-polar-form**
- **?@exr-complex-arithmetic-illustrate-addition-coordinate-system-parallelogram-rule**
- **?@exr-complex-sets-exercise**

- **?@exr-singularities-and-contour-integrals-consider-function-how-many-poles-does**

For the curious

- Exercise 4.4 — Stirling's approximation
- Exercise 5.3 — Pascal's identity
- Exercise 5.4 — Pascal's triangle and the Sierpinski triangle
- **?@exr-complex-vector-space-complex-numbers-real-vector-space** — Complex numbers as a real vector space
- **?@exr-eigenvalues-and-diagonalization-eigenvalue-decomposition**
- **?@exr-gram-schmidt-gram-schmidt-orthogonalization**
- **?@exr-singular-value-decomposition-singular-value-decomposition-compression-tool**
- **?@exr-dual-numbers-dual-numbers**
- **?@exr-complex-step-differentiation-complex-step-method**

Recommended exercises by topic

This chapter contains lists of mathematics exercises grouped by each quantum chemistry topic taught at the school. These exercises can be useful as a warm-up to the proper exercises given each day.

Coupled-cluster theory

No exercises selected.

Second quantization

Beginner

- Exercise 3.1 — Very basic sums
- Exercise 3.2 — Simplifying as a sum
- Exercise 3.4 — Reconstruct a vector
- Exercise 3.6 — The summation index is a dummy index
- Exercise 3.7 — Linearity of sums
- Exercise 3.8 — When something does not depend on the summation index
- Exercise 3.9 — Free and summed indices
- Exercise 3.10 — Shifting the summation index
- Exercise 3.11 — The Kronecker delta: basic exercise
- Exercise 5.1 — Basic binomial coefficients

Self-consistent field theory

No exercises selected.

Density functional theory

No exercises selected.

Response theory

No exercises selected.

Relativistic quantum chemistry

Beginner

- Exercise 3.12 — The Levi–Civita symbol: basic exercise

Molecular properties

No exercises selected.

Multiconfigurational methods

No exercises selected.

Part II

Mathematical foundations

1 Sets and functions

Sets and set notation appear many places in quantum chemistry. The goal of this chapter is to train you in the basic usage of sets and set notation.

1.1 Exercises: abstract sets and functions

Exercise 1.1 (Days of the week)**.** Write the days of the week as a set

Exercise 1.2 (Five first natural numbers)**.** Write the five first natural numbers using set notation as indicated in the lecture notes

Exercise 1.3 (Ordered pair)**.** An *ordered pair* (x, y) is a two-element "set" where the order matters. Two ordered pairs (x, y) and (u, v) are equal if and only if $x = u$ and $y = v$. From this it follows that $(x, y) = (y, x)$ if and only if $x = y$. A definition of an ordered pair as a *set* is

$$(x, y) = \{\{x\}, \{x, y\}\}.$$

Show that $(x, y) = (y, x)$ if and only if $x = y$ using the set theoretic definition.

Exercise 1.4 (Rational numbers as ordered pairs)**.** A nonnegative rational number p/q, $p, q \in \mathbb{N}$, $q > 0$, can be written as an ordered pair (p, q). Write down the rational numbers 1/4, 2/3, 0, 1/3 as ordered pairs using set theory, both for the pair and for each integer. (Strictly speaking we the number itself is an *equivalence class* of such pairs, i.e., $pn/qn = p/q$, so we must identify (p, q) and (np, nq). Ignore equivalence classes in this exercise.)

Exercise 1.5 (Cartesian product of two sets)**.** The *cartesian product* of two sets A and B is

$$A \times B = \{(a, b) \mid a \in A,\ b \in B\}.$$

Write down the cartesian product of $\{\heartsuit, \diamondsuit, \spadesuit, \clubsuit\}$ and $\{1, 2, 3\}$, using the set theoretic definition for the ordered pair. You can skip spelling out the set theoretic definition of the natural numbers.

Exercise 1.6 (Ordered pair as a two-element set)**.** Similar to the previous exercise, write down $\{1, 2, 3\} \times \{2, 3, 4\}$.

Exercise 1.7 (Function from one set to another set)**.** A *function* $f : A \to B$ from one set A to another set B is a rule that assigns to every $a \in A$ precisely one $b \in B$. In terms of set theory, a function $f : A \to B$ is a subset of $A \times B$, such that

- For all $a \in A$ there exists $b \in B$ such that $(a, b) \in f$.
- For all $a \in A$ and $b, b' \in B$, if $(a, b) \in f$ and $(a, b') \in f$, then $b = b'$.

Write down the function $f : \{1, 2, 3\} \to \{1, 2, 3\}$, $f(1) = 2$, $f(2) = 3$, $f(3) = 1$, using the set theoretic definition of ordered pairs and the cartesian product.

De Morgan's laws are useful when discussing subsets A, B of a larger set X. Recall that the complement of A relative to X is $A^{\complement} = X \setminus A$. De Morgan's laws state that:

- $(A \cup B)^{\complement} = A^{\complement} \cap B^{\complement}$.
- $(A \cap B)^{\complement} = A^{\complement} \cup B^{\complement}$.

Exercise 1.8 (De Morgan's laws)**.** Draw a picture, representing X as the whole sheet, A and B as overlapping shapes, e.g., circles. Label A, B, $A \cap B$, and $A \cup B$, making another drawing if necessary. Convince yourself that De Morgan's laws are correct.

Exercise 1.9 (De Morgan's laws 2)**.** Prove De Morgan's laws mathematically. Note that the complement operation acts like negation of truth, i.e., $a \in A^{\complement}$ if and only if $a \in X$ and $a \notin A$.

1.2 Exercises: Sets in quantum chemistry

Exercise 1.10 (Classically allowed region for Morse potential)**.** Figure Figure 1.1 shows the *Morse potential* - a common model potential for vibrational motion of a diatomic molecule:

$$V(r) = D_e \left(1 - e^{-a(r-r_e)}\right)^2 - D_e,$$

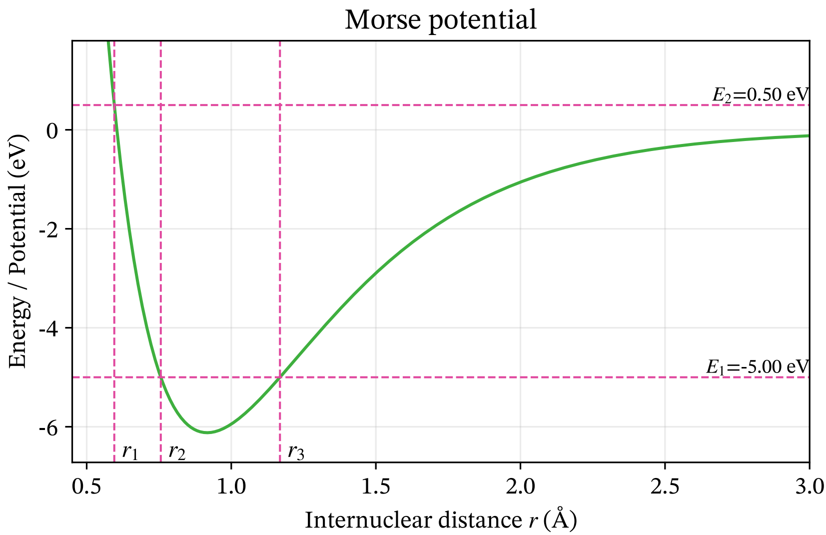

Figure 1.1: Morse potential

where $D_e > 0$ is the dissociation energy, $r_e > 0$ is the equilibrium bond length, and $a > 0$ is a parameter that determines the width of the potential well. The total energy of the molecule is denoted by $E \in \mathbb{R}$. The classically allowed region for the total energy E is the set S_E of all bond lengths r such that $V(r) \leq E$:

$$S_E = \{r \in \mathbb{R} \mid V(r) \leq E\}$$

a. Explain the notation $S_E = \{r \in \mathbb{R} \mid V(r) \leq E\}$ in words. In particular, what does the vertical bar $\mid$ mean?

b. Can you find a condition on E such that S_E is the empty set?

c. Can you find a condition on E such that S_E is the whole real line?

d. Based on the figure, which of the following statements are true?

- $S_{E_1} = (r_2, r_3)$
- $S_{E_1} = [r_2, r_3]$
- $S_{E_1} = \{r_2, r_3\}$

e. Based on the figure, which of the following statements are true?

- $S_{E_2} = (r_1, +\infty)$
- $S_{E_2} = [r_1, +\infty)$
- $S_{E_2} = \{r_1, r_2, r_3\}$

Exercise 1.11 (Molecules as sets)**.** In this exercise, a *molecule* is informally defined only by its chemical composition: the number of atoms of each element occurring in its molecular formula. Thus, H_2 and CO are two distinct molecules, while HCOOH and CO_2H_2 represent the same molecule, since both contain one C atom, two H atoms, and two O atoms.

We ignore molecular geometry, bonding, isotopes, charge, and all questions of chemical stability. In particular, any formal combination of atoms is deemed a molecule, whether or not such a species could exist physically.

We restrict our attention to the four elements C, H, O, and N. Let M denote the set of all molecules that can be constructed from these elements. A molecule must contain at least one atom.

One useful way of describing a molecule is by the ordered quadruple

$$(n_{\mathrm{C}}, n_{\mathrm{H}}, n_{\mathrm{O}}, n_{\mathrm{N}}), \tag{1.1}$$

where each n_X is the number of atoms of element X in the molecule. For example,

$$\mathrm{CH_4} \longleftrightarrow (1, 4, 0, 0),$$

and

$$\mathrm{N_2O} \longleftrightarrow (0, 0, 1, 2).$$

a. Explain why Equation 1.1 uniquely determines an element of M. Give 4-5 additional examples of molecules in terms of this representation.

b. Express M in terms of $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$. What is the cardinality of M? Is it finite, countably infinite, or uncountably infinite? Justify your answer.

c. Let

$$M_2 = \{m \in M \mid m \text{ contains exactly two atoms}\}.$$

Find $|M_2|$ and list all elements of M_2.

d. Let $S \subset M$ be the set

$$S = \{m \in M \mid m \text{ contains at least one H atom}\}.$$

Describe S in terms of the quadruple representation above. Is S finite, countably infinite, or uncountably infinite? Justify your answer.

d. Describe the complement $M \setminus S$. What is its cardinality?

e. Define

$$C = \{m \in M \mid m \text{ contains at least one C atom}\}.$$

Describe the following sets both in words and in terms of the quadruple representation:

$$C \cap S, \qquad C \cup S, \qquad C \setminus S.$$

Determine the cardinality of each set.

f. For $n \geq 1$, define

$$M_n = \{m \in M \mid m \text{ contains exactly } n \text{ atoms}\}.$$

Thus, a molecule in M_n satisfies

$$n_{\mathrm{C}} + n_{\mathrm{H}} + n_{\mathrm{O}} + n_{\mathrm{N}} = n.$$

Show that

$$|M_n| = \binom{n+3}{3}.$$

Check that your formula agrees with your answer to part **b**.

g. Explain why

$$M = \bigcup_{n=1}^{\infty} M_n.$$

Are the sets M_n pairwise disjoint? Use this decomposition to give another argument that M is countably infinite.

h. Show that M and S have the same cardinality. Explain why this is possible even though S is a proper subset of M.

2 Cardinality and countability

Cardinality compares sets through bijections. The exercises cover countable sets, Cantor's diagonal argument, and the cardinality of the continuum.

Recall that the *cardinality* $|S|$ of a set S is the number of elements in S. For finite sets, this is intuitive, but what about infinite sets?

One says that A and B have the same cardinality if there exists a bijection $f : A \to B$, i.e., f is one-to-one and onto. (Intuitively, you can draw lines between individual elements of A and B, only one line to/from each element. Precisely: Onto means, to every $b \in B$ there exists a $a \in A$ such that $f(a) = b$. One-to-one: $f(a) = f(a')$ implies $a = a'$.) One can think of the bijection as labelling each $b \in B$ by exactly one element $a \in A$, and that all labels from A are used up.

Let $\mathbb{N}_n = \{1, 2, \cdots, n\}$. This set has cardinality n.

The set $\mathbb{N}$ has an infinite number of elements. By definition, $|\mathbb{N}| = \aleph_0$ ("aleph-nought").

2.1 Exercises

Exercise 2.1 (Countability of the integers)**.** Show that $|\mathbb{Z}| = \aleph_0$ by explicitly constructing a bijection $f : \mathbb{N} \to \mathbb{Z}$.

Exercise 2.2 (Countability of the natural number plane)**.** Show that $|\mathbb{N} \times \mathbb{N}| = \aleph_0$. Hint: Draw a picture of $\mathbb{N} \times \mathbb{N}$, and try to draw a line through all the points in this set. How does this show the existence of a bijection between $\mathbb{N}$ and $\mathbb{N} \times \mathbb{N}$?

Exercise 2.3 (Countability of the rational numbers)**.** Show that $|\mathbb{Q}| = \aleph_0$.

Exercise 2.4 (Interval vs. real numbers)**.** Show that the interval $I = (-1, 1)$ has the same cardinality as $\mathbb{R}$. You must construct a function $f : (-1, 1) \to \mathbb{R}$ that is a bijection.

Exercise 2.5 (Uncountability of the real numbers)**.** In this exercise, we show that $|\mathbb{R}| > \aleph_0$. The cardinality would be $\aleph_0$ if we could write a list of the real numbers. It suffices to find a list of the numbers in $I = [0, 1[$, i.e., infinite decimal expansions $0.d_1d_2d_3\cdots$, with $d_i \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

We will look for a contradiction. Consider a list $x_1, x_2, x_3, \ldots$ of all real numbers in $[0, 1[$. This is essentially an infinite matrix of digits. Can you find a real number $x \in [0, 1]$ which is *not* in this list, i.e., $x \neq x_j$ for all j? Hint: consider the diagonal of the matrix.

The cardinality of $\mathbb{R}$, "the continuum", is written $2^{\aleph_0}$.

Exercise 2.6 (Constructing a one-to-one map)**.** Show that $|[0, 1[\times[0, 1[| = |[0, 1[|$, by constructing a one-to-one map. Hint: Use decimal expansions. Conclude that also $|\mathbb{R}^2| = |\mathbb{C}| = |\mathbb{R}|$.

3 Summing over indices

Summing several things in a single expression occurs frequently in mathematics, and in particular in applications to physics and chemistry. The general syntax for a summation over an index is:

$$\sum_{i \in I} f(i),$$

where I is some countable set of indices, i is the index of summation, and $f(i)$ is the function/quantity being summed. The set I can be finite or infinite, and the index i can be a number, a vector, or even a more complicated object.

The most basic and common case is a sum over a finite or infinite set of integers, which is usually written as

$$\sum_{i=a}^{b} f(i)$$

where i is the index of summation, a is the lower limit of the index i, b is the upper limit, and $f(i)$ is the function/quantity being summed.

Here are some concrete examples:

1. A sum over a finite number of terms:

$$\sum_{i=1}^{5} i = 1 + 2 + 3 + 4 + 5$$

2. An infinite series:

$$\sum_{n=0}^{\infty} \frac{1}{2^n} = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$$

3. A series expansion. In this example, the exponential function is expressed as a Taylor series, summing over all non-negative integers:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots$$

4. Vector notation. For example, a vector $\mathbf{r} \in \mathbb{R}^3$ can be expressed as a sum over its components:
$$\sum_{i=1}^{3} r_i \mathbf{e}_i = \mathbf{r}$$

5. A matrix-vector product expressed as a sum over indices: The i-th component of the product of a matrix A and a vector $\mathbf{x}$ is given by the sum over the products of the corresponding elements of the matrix and vector:
$$(A\mathbf{x})_i = \sum_{j=1}^{n} A_{ij} x_j$$

3.0.1 Manipulating sums

The index in a sum is called a **summation index** or **dummy index**. Its name has no significance. For example,

$$\sum_{i=1}^{n} a_i = \sum_{j=1}^{n} a_j.$$

The index may therefore be renamed, as long as it is renamed consistently within the sum.

Summation is **linear**. Sums can be split or combined,

$$\sum_{i=1}^{n} (a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i,$$

and factors that do not depend on the summation index can be taken outside the sum,

$$\sum_{i=1}^{n} c a_i = c \sum_{i=1}^{n} a_i,$$

where it is important that c is independent of i.

Expressions may contain both summed and unsummed indices. An index that is not summed over is called a **free index**. For example, in the matrix-vector product

$$y_i = \sum_{j=1}^{n} A_{ij} x_j,$$

j is a dummy index, while i is a free index. The equation therefore represents one equation for each possible value of i.

3.0.2 Special symbols used in summation notation

The **Kronecker delta** δ_{ij} is a special symbol that is often used in summation notation. It is defined as

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Sometimes, if confusion can arise one inserts a comma between the two indices, writing $\delta_{i,j}$ instead of δ_{ij}.

The Kronecker delta forms the matrix elements of the identity matrix.

The **Levi-Civita symbol** ϵ_{ijk} is another special symbol that is often used in summation notation in the three-dimensional case, i.e., where $i, j, k \in \{1, 2, 3\}$. It is defined as

$$\epsilon_{ijk} = \begin{cases} 1 & \text{if } (i, j, k) \text{ is an even permutation of } (1, 2, 3), \\ -1 & \text{if } (i, j, k) \text{ is an odd permutation of } (1, 2, 3), \\ 0 & \text{if any two indices are equal.} \end{cases}$$

3.1 Exercises: Basic summation

These exercises are at a very basic level. If you are already comfortable with summation notation, you may skip them.

Exercise 3.1 (Very basic sums). Expand the following sums and calculate their values:

a) $\sum_{i=1}^{4} i$

b) $\sum_{i=1}^{4} i^2$

c) $\sum_{k=0}^{3} 2^k$

Exercise 3.2 (Simplifying as a sum). Write each expression using summation notation:

a) $x_1 + x_2 + x_3 + x_4 + x_5$

b) $1 + q + q^2 + q^3 + q^4$

c) $a_1 b_1 + a_2 b_2 + a_3 b_3$

Exercise 3.3 (Estimating $\sqrt{e}$ via a Taylor series). Use the first four terms of the series for e^x to estimate $\sqrt{e}$:

$$\sqrt{e} = e^{0.5} \approx \sum_{n=0}^{3} \frac{(0.5)^n}{n!}.$$

Write out the four terms and add them. The estimate is not especially accurate; the point is to see how the index n controls both the power and the factorial.

Exercise 3.4 (Reconstruct a vector). The components of a vector are $r_1 = 2$, $r_2 = -1$, and $r_3 = 4$. Let $\mathbf{e}_i$ be the standard basis in $\mathbb{R}^3$.

a) Write out $\sum_{i=1}^{3} r_i \mathbf{e}_i$ term by term.

b) Gather the result as a column vector

Exercise 3.5 (Summing in a matrix–vector product). Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 0 & 3 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}.$$

Use

$$(A\mathbf{x})_i = \sum_{j=1}^{3} A_{ij} x_j$$

to write out and calculate $(A\mathbf{x})_1$ and $(A\mathbf{x})_2$. Collect the two answers in a column vector.

Exercise 3.6 (The summation index is a dummy index)**.** Consider

$$S = \sum_{i=1}^{4} i^2.$$

a. Write out the sum explicitly and calculate its value.

b. Write out and calculate

$$\sum_{j=1}^{4} j^2.$$

c. Explain why

$$\sum_{i=1}^{4} i^2 = \sum_{j=1}^{4} j^2.$$

d. Rewrite

$$\sum_{k=1}^{5} a_k b_k$$

using p as the summation index.

Exercise 3.7 (Linearity of sums)**.** Let a_i and b_i, $i = 1, 2, 3$ be two sets of numbers and let c be a constant, i.e., just a number.

a. Write out both sides of

$$\sum_{i=1}^{3}(a_i + b_i) = \sum_{i=1}^{3} a_i + \sum_{i=1}^{3} b_i$$

and verify that they are equal.

b. Write out both sides of

$$\sum_{i=1}^{3} ca_i = c\sum_{i=1}^{3} a_i$$

and verify that they are equal.

c. Use these properties to rewrite

$$\sum_{i=1}^{3}(2a_i - 3b_i)$$

as two separate sums (i.e., each has a single summation sign).

Exercise 3.8 (When something does not depend on the summation index)**.** Let x be a number independent of the summation index i.

a. Simplify
$$\sum_{i=1}^{5} x.$$

b. Simplify
$$\sum_{i=1}^{5} ix.$$

c. Explain why x can be taken outside the sum in
$$\sum_{i=1}^{n} ix = x \sum_{i=1}^{n} i.$$

Exercise 3.9 (Free and summed indices)**.** Consider the following expression for the i-th component of a vector $\mathbf{y}$, being the product of a matrix A and a vector $\mathbf{x}$:
$$y_i = \sum_{j=1}^{3} A_{ij} x_j.$$

a. Which index is summed over?

b. Which index is not summed over?

c. For $i = 2$, write the right-hand side explicitly.

d. If A has four rows and three columns, how many components does $\mathbf{y}$ have?

e. Explain why replacing j by k,
$$y_i = \sum_{k=1}^{3} A_{ik} x_k,$$
does not change the equation.

Exercise 3.10 (Shifting the summation index)**.** Consider
$$S = \sum_{i=1}^{4} a_i.$$

a. Write out S explicitly.

b. Show that the same sum can be written as

$$S = \sum_{j=0}^{3} a_{j+1}$$

by writing out the terms.

c. Rewrite

$$\sum_{i=2}^{5} a_i$$

as a sum whose index starts at $j = 0$.

d. Rewrite

$$\sum_{i=0}^{n-1} a_{i+1}$$

as a sum with index j running from 1 to n.

Exercise 3.11 (The Kronecker delta: basic exercise). The **Kronecker delta** is defined in the introduction to this chapter. It is a special symbol that is often used in summation notation.

a. Evaluate δ_{11}, δ_{12}, δ_{23}, and δ_{33}.

b. Let $a_1 = 2$, $a_2 = 5$, and $a_3 = 7$. Write out all terms in the sum

$$\sum_{j=1}^{3} \delta_{2,j} a_j$$

and evaluate it.

c. Repeat (b) for

$$\sum_{j=1}^{3} \delta_{1j} a_j \qquad \text{and} \qquad \sum_{j=1}^{3} \delta_{3j} a_j.$$

d. Based on your results, simplify the general expression

$$\sum_{j=1}^{n} \delta_{ij} a_j.$$

e. In the expression in (d), which index is a dummy index and which is a free index? What values can the free index take?

Exercise 3.12 (The Levi–Civita symbol: basic exercise)**.** The **Levi–Civita symbol** ε_{ijk} is defined in the introduction to this chapter.

a. Evaluate ε_{123}, ε_{132}, ε_{231}, ε_{321}, and ε_{223}.

b. Find all triples (i, j, k) for which $\varepsilon_{ijk} = +1$.

c. Find all triples (i, j, k) for which $\varepsilon_{ijk} = -1$.

d. What happens to ε_{ijk} if two indices are interchanged? Illustrate with two examples.

e. Let a_1, a_2, and a_3 be the components of a vector. Write out the sum

$$\sum_{j=1}^{3}\sum_{k=1}^{3} \varepsilon_{1jk} a_j b_k$$

and remove all terms that vanish because of the Levi–Civita symbol.

Exercise 3.13 (A molecular orbital in an atomic-orbital basis)**.** In quantum chemistry, a molecular orbital $\psi_p(\mathbf{r})$ is commonly expanded in a finite set of atomic-orbital basis functions $\chi_\mu(\mathbf{r})$,

$$\psi_p(\mathbf{r}) = \sum_{\mu=1}^{K} C_{\mu p} \chi_\mu(\mathbf{r}).$$

Here, μ labels the basis functions, while p labels the molecular orbital.

Consider three basis functions and the coefficient matrix

$$C = \begin{bmatrix} 0.7 & 0.5 \\ 0.7 & -0.5 \\ 0.1 & 0.7 \end{bmatrix}.$$

a. In the expression for $\psi_p(\mathbf{r})$, identify the dummy index and the free index.

b. Write $\psi_1(\mathbf{r})$ explicitly as a linear combination of $\chi_1(\mathbf{r})$, $\chi_2(\mathbf{r})$, and $\chi_3(\mathbf{r})$, using the matrix elements of C.

c. Do the same for $\psi_2(\mathbf{r})$.

d. Suppose that at a particular point $\mathbf{r}_0$,

$$\chi_1(\mathbf{r}_0) = 1.0, \qquad \chi_2(\mathbf{r}_0) = 0.5, \qquad \chi_3(\mathbf{r}_0) = 0.2.$$

Calculate $\psi_1(\mathbf{r}_0)$ and $\psi_2(\mathbf{r}_0)$.

e. Explain why the summation index μ could be replaced by another symbol, such as i, without changing the meaning of the expression.

4 Combinatorics

Combinatorics is the part of mathematics concerned with *counting* in a broad sense, for example counting the number of ways to arrange a set of objects or the number of combinations that can be made from a set of objects. It is a fundamental part of discrete mathematics and has applications in many areas of mathematics, computer science, chemistry, and physics.

4.1 Exercises: Factorials

The *factorial* of a non-negative integer n is defined as the product of all positive integers less than or equal to n:

$$n! = n \cdot (n-1) \cdot (n-2) \cdots 3 \cdot 2 \cdot 1, \quad n \geq 1.$$

It is also defined that $0! = 1$. The factorial appears in many mathematical contexts, including series expansions, and of course combinatorics. Its basic use is the following:

> **Factorial and counting**
>
> Given a set of n distinct objects, the number of ways to arrange them in a sequence (i.e., the number of permutations) is $n!$.

In one exercise, you will need the following approximation for large n, known as Stirling's approximation:

> **Stirling's approximation**
>
> For large n, the factorial $n!$ can be approximated as

$$n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n.$$

In calculations, especially numerical calculations, it is often convenient to use the logarithm of the factorial. Using Stirling's approximation, we have

$$\log(n!) \approx \frac{1}{2}\log(2\pi n) + n\log(n) - n.$$

Exercise 4.1 (Basic factorials)**.** Compute the factorials 0!, 1!, 2!, 3!, 4!, and 5!.

Exercise 4.2 (Lunch line)**.** 42 people stand in a lunch line. How many different lunch line arrangements with 42 people are possible?

Exercise 4.3 (Chemist's shelf)**.** Compute the number of arrangements for the following scenarios:

a. A chemist has 10 different chemical samples. How many ways can the chemist arrange the samples on a shelf?
b. A chemist has 10 different chemical samples. 4 of the samples are liquids, and the other 6 are solids. How many ways can the chemist arrange the samples on a shelf if all the liquids must be together?

Exercise 4.4 (Stirling's approximation)**.** We study Stirling's approximation for the factorial.

a. Use Stirling's approximation to estimate 5!. Compare your result with the exact result

$$5! = 120.$$

b. Use Stirling's approximation to estimate 10!. Compare again with the exact number

$$10! = 3\,628\,800.$$

c. The relative error of an approximation A to an exact value x is

$$\frac{|A - x|}{x}.$$

Calculate the relative error of Stirling's approximation for 5! and 10!. For which value of n is the approximation better?

d. Use Stirling's approximation to estimate 50!. The exact value is approximately

$$50! \approx 3.041 \times 10^{64}.$$

e. Based on your results above, what happens to the accuracy of Stirling's approximation as n increases?

5 Exercises: Binomial coefficients and Pascal's triangle

The *binomial coefficient* $\binom{n}{k}$ is defined as the number of ways to choose k elements from a set of n elements, without regard to the order of selection. It is given by the formula:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!},$$

Pascal's triangle is a triangular array of numbers where each number is the sum of the two numbers directly above it:

$$
\begin{array}{ccccccccc}
 & & & & 1 & & & & \\
 & & & 1 & & 1 & & & \\
 & & 1 & & 2 & & 1 & & \\
 & 1 & & 3 & & 3 & & 1 & \\
1 & & 4 & & 6 & & 4 & & 1 \\
 & & & & \vdots & & & &
\end{array}
\qquad
\begin{array}{r}
(5.1)\\ (5.2)\\ (5.3)\\ (5.4)\\ (5.5)\\ (5.6)
\end{array}
$$

Fact: The n-th row of Pascal's triangle contains the binomial coefficients $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$.

Exercise 5.1 (Basic binomial coefficients)**.**

a. Compute the binomial coefficients $\binom{5}{2}$, $\binom{6}{3}$, and $\binom{7}{4}$.
b. In Norwegian Lotto, 7 numbers are drawn from a set of 34 numbers. How many different combinations of 7 numbers can be drawn?

Exercise 5.2 (Pascal's triangle)**.** Compute the first 6 rows of Pascal's triangle. Verify that the numbers in the n-th row correspond to the binomial coefficients $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$.

Exercise 5.3 (Pascal's identity)**.** The binomial coefficients satisfy **Pascal's identity**,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

a. Explain why this relation is equivalent to the construction of Pascal's triangle, where each number is the sum of the two numbers directly above it.

b. Write each binomial coefficient on the right-hand side in terms of factorials.

c. Put the two terms over a common denominator and simplify the expression.

d. Show that the resulting expression is

$$\frac{n!}{k!(n-k)!},$$

and hence prove Pascal's identity.

e. Give a combinatorial explanation of Pascal's identity. *Hint:* Consider choosing k objects from a set of n objects. Pick one particular object and separate the choices into two cases: those that contain this object and those that do not.

Exercise 5.4 (Pascal's triangle and the Sierpinski triangle)**.** Write a computer program to generate the first 2^n rows of Pascal's triangle, where $n = 1, 2, \dots$. Create a $2^n \times 2^n$ lower triangular matrix T, where $T_{k,n} = 0$ if $\binom{n}{k}$ is even, and $T_{k,n} = 1$ if the coefficient is odd, for $k \le n$ (indexing from zero). Visualize the matrix. What pattern do you observe? This pattern is known as the Sierpinski triangle.

If you use Python and sympy, you can generate binomial factors with the function `sympy.binomial(n, k)`. Check for parity using the modulo operator `%`. For example, `sympy.binomial(n, k) % 2` will return `0` for even numbers and `1` for odd numbers.

Part III

Linear algebra

6 Vectors and matrices

The vector space $\mathbb{R}^n$ consists of all column vectors

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \qquad x_i \in \mathbb{R},$$

with the usual vector addition and scalar multiplication. Similarly, $\mathbb{C}^n$ consists of column vectors with complex components. More generally, a vector space V over a field $\mathbb{F}$ is a set equipped with vector addition and scalar multiplication such that, for all $x, y, z \in V$ and $\alpha, \beta \in \mathbb{F}$,

1. **Commutativity of addition:** $x + y = y + x$.
2. **Associativity of addition:** $x + (y + z) = (x + y) + z$.
3. **Additive identity:** There exists a zero vector $0 \in V$ such that $x + 0 = x$.
4. **Additive inverse:** For every $x \in V$, there exists $-x \in V$ such that $x + (-x) = 0$.
5. **Distributivity over vector addition:** $\alpha(x + y) = \alpha x + \alpha y$.
6. **Distributivity over scalar addition:** $(\alpha + \beta)x = \alpha x + \beta x$.
7. **Associativity of scalar multiplication:** $(\alpha\beta)x = \alpha(\beta x)$.
8. **Scalar identity:** $1x = x$.

For a matrix $A \in \mathbb{R}^{m \times n}$ and a column vector $x \in \mathbb{R}^n$, the matrix-vector product $y = Ax \in \mathbb{R}^m$ is defined componentwise by

$$y_i = \sum_{j=1}^{n} A_{ij} x_j.$$

For matrices $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$, the matrix product $C = AB \in \mathbb{R}^{m\times p}$ is defined by

$$C_{ij} = \sum_{k=1}^{n} A_{ik}B_{kj}.$$

The same definitions of matrix-vector and matrix-matrix multiplication apply over $\mathbb{C}$.

6.1 Exercises: Basic vectors and matrix calculations

Exercise 6.1 (Basic vector operations)**.** Consider the vectors

$$x = (1, 2, -1), \qquad y = (3, -1, 2), \qquad z = (0, 2, 1)$$

in the vector space $\mathbb{R}^3$.

a. Compute $x + y$.
b. Compute $x - y$.
c. Compute $2x$ and $-3y$.
d. Compute the linear combination $2x + 3y$.
e. Compute $x + 2y - z$.
f. Find numbers α and β such that $\alpha x + \beta y = (7, 0, 3)$.
g. Verify by direct calculation that $2(x + y) = 2x + 2y$. Which vector-space axiom does this illustrate?
h. Is $(1, 2)$ an element of this vector space? Is $(0, 0, 0)$ an element? Explain briefly.

Exercise 6.2 (Matrix-vector products)**.** Consider the matrices and vectors

$$A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{pmatrix}, \qquad x = \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \qquad y = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}.$$

a. Compute Ax.

b. Compute By.

c. Compute $A(2x)$ and $2(Ax)$. Verify that they are equal.

d. Can the products Ay and Bx be formed? Explain your answer in terms of the dimensions of the matrices and vectors.

Exercise 6.3 (Matrix-matrix products)**.** Consider the matrices

$$A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 0 \\ -1 & 4 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 2 \end{pmatrix}.$$

a. Compute AB.

b. Compute BA. Is $AB = BA$?

c. Compute AC. What are the dimensions of the resulting matrix?

d. Can the product CA be formed? Can the product BC be formed? Explain your answers using the dimensions of the matrices.

Exercise 6.4 (Linear independence and basis)**.** We work in the space $V = \mathbb{R}^2$, plane vectors. We define two vectors,

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

a. Show by elementary means that the two vectors are linearly independent.
b. Why do they form a basis for V?
c. Make a 2D drawing or plot of them as arrows from the origin.

Exercise 6.5 (Gaussian elimination and vector decomposition)**.** Let

$$\mathbf{v} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Using Gaussian elimination, compute the coefficients x_1 and x_2 such that

$$\mathbf{v} = x_1\mathbf{b}_1 + x_2\mathbf{b}_2.$$

Illustrate this decomposition in the graph from the previous exercise, using the parallelogram rule: The sum of two vectors is obtained by placing the origin of one at the end of the second, or vice versa.

Exercise 6.6 (Office coordinates). A person walks from home. First, they walk to the subway station. This is 1 km north and 500 m east as the crow flies. The subway takes the person to the city centre, which 3 km east and 2 km south of the subway station, as the crow flies. The person then walks 200 m south and 400 m west to their office.

What are the coordinates of the office, relative to the person's home?

What is the distance to home, as the crow flies?

Exercise 6.7 (Boat sailing on the ocean). A boat is sailing on the ocean, at a constant speed 20 km/h relative to the water surface, in the northeast direction. However, the water surface is moving too, at a constant velocity. After two hours, the boat is located 20 km north and 10 km east of the starting point. What is the speed and direction of ocean drift?

Exercise 6.8 (An exercise by Robert Beezer). A three-digit number has two properties. The tens-digit and the ones-digit add up to 5. If the number is written with the digits in the reverse order, and then subtracted from the original number, the result is 792. Use a system of equations to find all of the three-digit numbers with these properties.

6.2 Exercises: More vectors and matrices

Exercise 6.9 (Compute the action of a matrix on vectors). Let A be the $n \times n$ matrix defined by $A_{i,i+1} = i^2$ for $1 \leq i < n$, and zero otherwise. Compute the action of A and A^T on the standard basis for $\mathbb{R}^n$. Compute the action of A and A^T on an arbitrary vector $\mathbf{x} \in \mathbb{R}^n$.

Exercise 6.10 (Matrix-matrix products, rows and columns). Let $A \in M(n, m, \mathbb{F})$, $B \in M(m, o, \mathbb{F})$. Write B in terms of column vectors, $B = [\mathbf{b}_1, \cdots, \mathbf{b}_o]$. Prove that

$$AB = [A\mathbf{b}_1, \cdots, A\mathbf{b}_o],$$

i.e., A acts on each individual column of B. Similarly, show that, if we write A in terms of its *row vectors*,

$$A = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \cdots \\ \mathbf{a}_n^T \end{bmatrix},$$

then

$$AB = \begin{bmatrix} \mathbf{a}_1^T B \\ \mathbf{a}_2^T B \\ \cdots \\ \mathbf{a}_n^T B \end{bmatrix}.$$

Exercise 6.11. Show that each column of $C = AB$ is a linear combination of the columns of A with coefficients determined by B,

$$\mathbf{c}_i = \sum_j \mathbf{a}_j B_{ji}.$$

Show a similar result for the rows of C.